

Yearwork Dynamics of Mechanical Systems

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1 First Question

To make the subdivision in finite elements of the system I followed the two main criteria we've seen in the theory:

1. Adding a node in every intersection
2. Considering a maximum length of finite elements so that each of them works in the quasi-static region.

The 2) criteria requires some clarifications. In the system we can identify basically three types of finite elements: horizontal, vertical and diagonal. We can identify a maximum length for each category of finite elements and see if we can keep the subdivision obtained by applying just the first criteria or we need a further subdivision since the finite elements exceed the maximum length that guarantees our assumption to be correct.

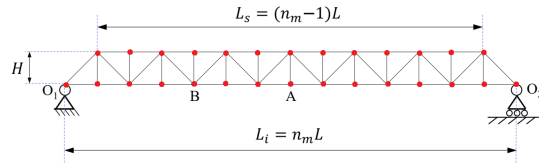


Figure 1: Subdivision obtained applying first criteria

1.1 Calculation of $L_{\max}$

The general condition we have to impose for each finite element k is:

$$\omega_k \ll \Omega_{\max}$$

Where:

- $\Omega_{\max}$ is the maximum exciting frequency which in our case is 30 Hz (from data).
- ω_k is the smallest bending natural frequency of the k -th finite element

By considering the boundary condition for pinned-pinned beams and the modal shapes for bending and we obtain the following general formula:

$$\left(\frac{\pi}{L_{kmax}} \right)^2 \sqrt{\frac{EI_k}{\rho A}} = \max f$$

And explicitating L_{kmax} :

$$L_{kmax} = \sqrt{\frac{\pi}{\sqrt{45 \cdot 2\pi} \cdot \sqrt{\frac{m_k}{E \cdot I_k}}}}$$

We used 45 Hz as the maxf value (by applying the safety coefficient of 1.5), so that we can write an equation (not an inequality) and find an exact maximum value for the length of the finite elements. Clearly the only thing that changes for the different categories of finite elements in the system is I_k and m_k .

1.1.1 Horizontal Finite Elements

$$\begin{aligned} m_K &= A \cdot \rho = 115.5 \cdot 10^{-4} \text{ m}^2 \cdot 7800 \text{ kg/m}^3 = 90.09 \text{ kg/m} \\ J &= 48200 \text{ cm}^4 = 48200 \cdot 10^{-8} \text{ m}^4 = 4.82 \cdot 10^{-4} \text{ m}^4 \\ L_K &= 6.05 \text{ m} \end{aligned}$$

In our subdivision each horizontal finite element has a length of 5 meters so we can keep them in the same subdivision as before since:

$$5 < 6.05$$

1.1.2 Vertical Finite Elements

$$\begin{aligned} m_K &= 76.84 \cdot 10^{-4} \text{ m}^2 \cdot 7800 \text{ kg/m}^3 = 59.9 \text{ kg/m} \\ J &= 7763 \text{ cm}^4 = 7763 \cdot 10^{-8} \text{ m}^4 = 0.776 \cdot 10^{-4} \text{ m}^4 \\ L_K &= 4.24 \text{ m} \end{aligned}$$

In our subdivision each vertical finite element has a length of 4 meters so we can keep them in the same subdivision as before since:

$$4 < 4.24$$

1.1.3 Diagonal Finite Elements

$$\begin{aligned} m_K &= 112.5 \cdot 10^{-4} \text{ m}^2 \cdot 7800 \text{ kg/m}^3 = 87.75 \text{ kg/m} \\ J &= 18260 \text{ cm}^4 = 18260 \cdot 10^{-8} \text{ m}^4 = 1.82 \cdot 10^{-4} \text{ m}^4 \\ L_K &= 4.48 \text{ m} \end{aligned}$$

In our subdivision each diagonal finite element has a length of 7.07 meters:

$$6.4 > 4.48$$

So we must change our initial subdivision and we decide to split our previous diagonal finite element in a half. The final subdivision we obtain is the following one (with numeration of the nodal sections):

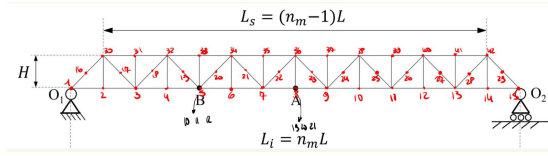


Figure 2: Final subdivision with enumeration of nodal sections

2 Second Question

I started by simply writing the text file to upload the structure and the subdivision we previously created. Then I've founded the mode of vibrations by exploiting the dedicated software functionality only considering the modes inside the frequency range of interest:

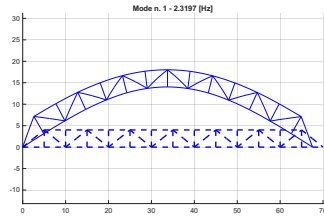


Figure 3: 1st mode of vibration

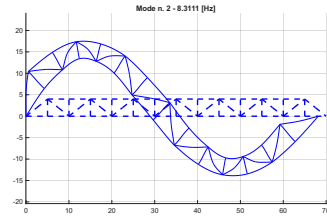


Figure 4: 2nd mode of vibration

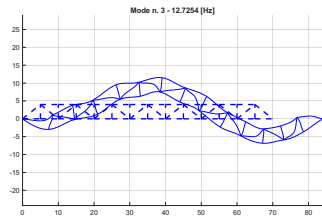


Figure 5: 3rd mode of vibration

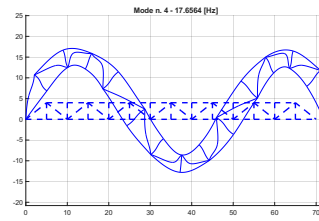


Figure 6: 4th mode of vibration

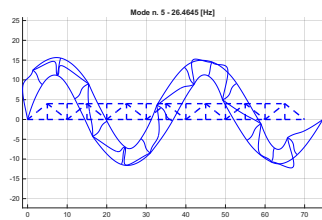


Figure 7: 5th mode of vibration

3 Third Question

3.1 Results

Even for the third question I simply used the dedicated functionality of the software considering always the magnitude of the force equal to 1. The frequency response functions describing the **vertical displacement** on point A and B obtained by applying a vertical force on point A are the following:

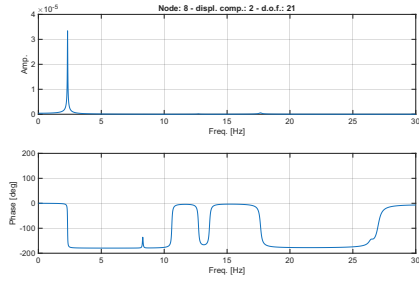


Figure 8: Displacement point A

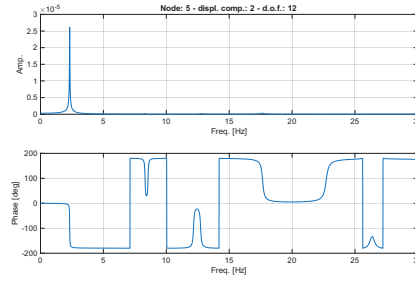


Figure 9: Displacement point B

The frequency response functions describing the **vertical acceleration** on point A and B obtained by applying a vertical force on point B are the following:

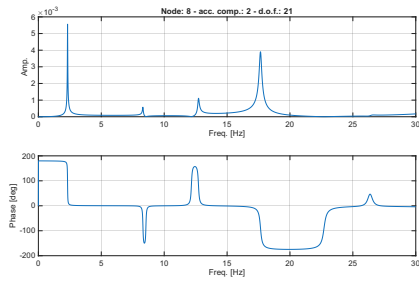


Figure 10: Acceleration point A

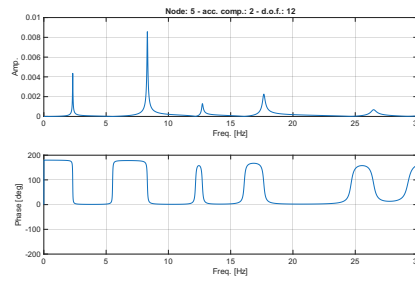


Figure 11: Acceleration point B

4 Fourth Question

4.1 Procedure

Here are listed the general step of the procedure followed:

1. The seismic motion at the supports is read from a text file as time histories of imposed displacements $y_{O1}(t)$ and $y_{O2}(t)$, and the sampling parameters are defined.
2. The imposed displacement signals are transformed into the frequency domain by means of the Fast Fourier Transform (FFT), obtaining the complex spectra $Y_{O1}(\omega)$ and $Y_{O2}(\omega)$.

3. The finite element model of the bridge is loaded and the mass, damping, and stiffness matrices are partitioned into free (F) and constrained (C) degrees of freedom.
4. The dynamic problem is formulated with kinematic excitation, assuming the support displacements as known inputs.
5. For each angular frequency ω , the equivalent dynamic forces induced by the imposed support motion are computed in the frequency domain.
6. The dynamic equilibrium equations are solved in the frequency domain using the frequency response function (FRF) to obtain the complex displacements of the free degrees of freedom.
7. The displacements at points A and B are extracted, and the corresponding accelerations are computed in the frequency domain.
8. The time histories of displacements and accelerations are reconstructed by applying the inverse Fast Fourier Transform (IFFT).

4.2 Results

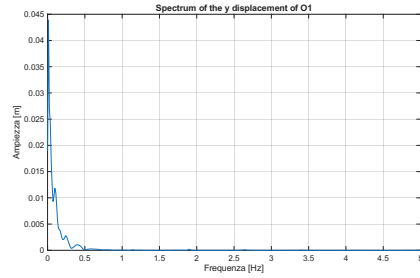


Figure 12: Figura 1

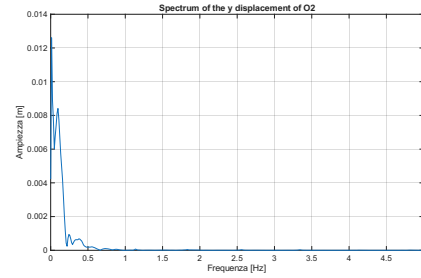


Figure 13: Figura 2

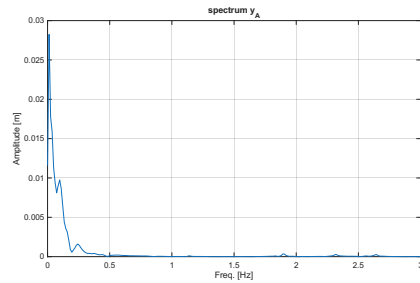


Figure 14: Figura 1

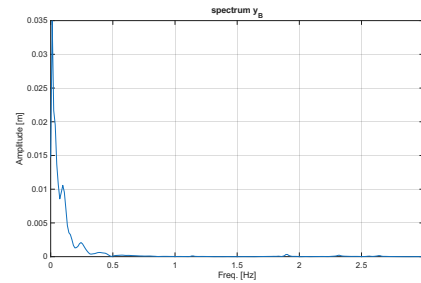


Figure 15: Figura 2

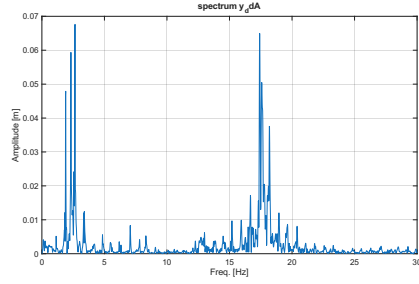


Figure 16: Figura 1

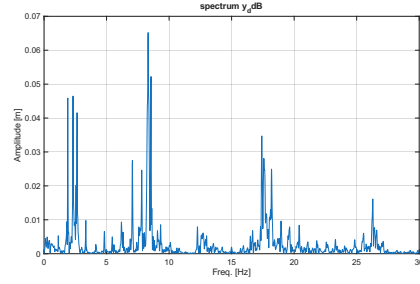


Figure 17: Figura 2

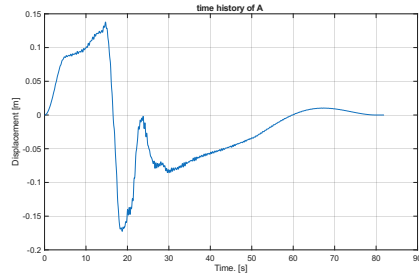


Figure 18: Figura 1

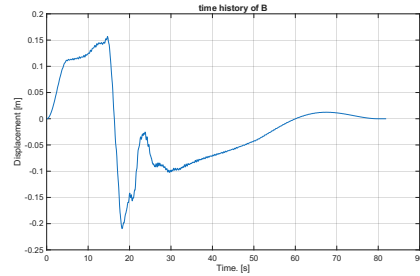


Figure 19: Figura 2

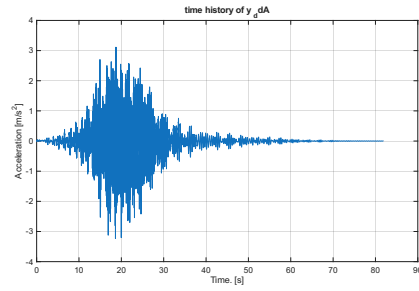


Figure 20: Figura 1

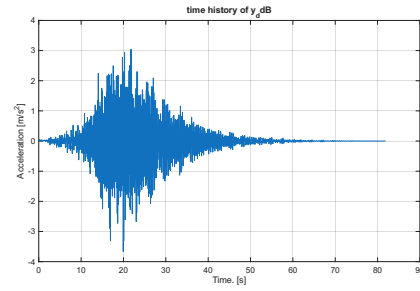


Figure 21: Figura 2

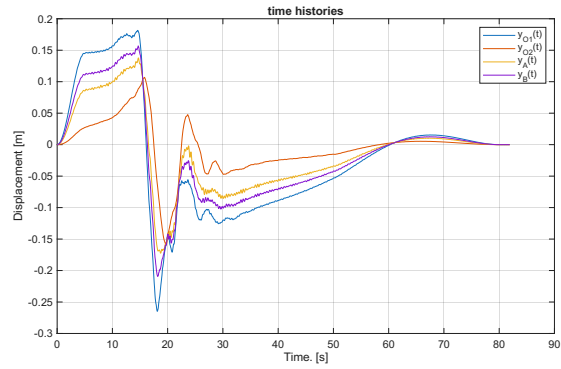


Figure 22

5 Fifth Question

Assuming the train as a sequence of **ideal moving impulses** ($L = 0$), the forcing on the bridge can be represented as a **periodic train of Dirac delta loads**. The fundamental frequency of the train is determined solely by the spacing D between consecutive loads:

$$f_0 = \frac{V}{D}.$$

The spectrum contains all integer multiples of the fundamental, i.e., $f_n = n f_0$. A resonance occurs whenever a harmonic coincides with a natural frequency of the bridge. Considering the **fundamental harmonic only** ($n = 1$), the train velocities required to excite the natural frequencies of the bridge modes

$$f_{\text{modes}} = [2.3197, 8.311, 12.72, 17.65, 26.46] \text{ Hz}$$

are calculated as

$$V = f_{\text{mode}} \cdot D,$$

which yields:

Mode	f_{mode} [Hz]	V [m/s]	V [km/h]
1	2.3197	60.31	217.1
2	8.311	215.99	777.6
3	12.72	330.72	1190.6
4	17.65	458.9	1652.0
5	26.46	687.96	2476.7

For the **first mode**, considering also the **second harmonic** ($n = 2$), the required velocities are:

$$V = \frac{f_1^{\text{mode}}}{n} \cdot D,$$

yielding:

Harmonic	V [m/s]	V [km/h]
1	60.31	217.1
2	30.16	108.6

In practice, the **second harmonic already has a significantly smaller amplitude** than the first. This is because in reality $L \neq 0$, so the loads are not ideal impulses and not all harmonics have the same amplitude. For a rectangular pulse of duration L/V , the amplitude of the n -th harmonic is proportional to

$$|F_n| \propto \frac{\sin(n\pi\delta)}{n\pi}, \quad \delta = \frac{L}{L + D},$$

which clearly decreases for higher n . Therefore, considering **only the fundamental harmonic** is a **conservative and sufficient approximation** for evaluating the resonance of the bridge.

6 Sixth Question

The new structure I considered to decrease point B acceleration is the following:

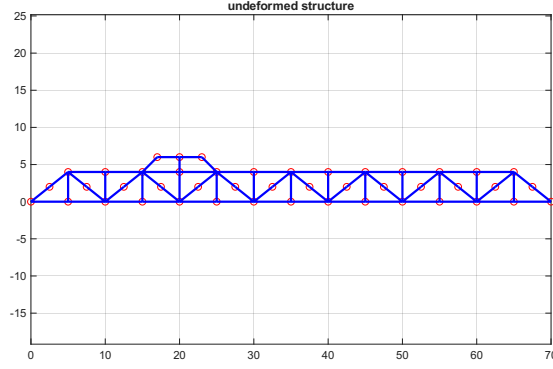


Figure 23

The total mass of the old structure can be estimated as:

$$M_{\text{tot}} = \rho \left((2n_m - 1)A_x + 14\sqrt{5^2 + 4^2} A_d + 13 \cdot 4 A_y \right) = 22695$$

so, to respect the specific given in question six, the mass of all the additional element introduced by the new design must respect this condition:

$$M_{\text{add}} < M_{\text{tot}} \cdot \frac{5}{100} = 1135\text{kg}$$

To modify the basic structure I used 1 horizontal beam, 1 diagonal beam splitted in 2 half and one half of a vertical beam. The mass of all this elements is **1132 kg** and so the condition stated above is respected.

Let Y_0 be the original acceleration of node B and Y_1 the acceleration after the change in the structure is applied. The percentage reduction in acceleration can be calculated as:

$$\Delta Y\% = \frac{Y_0 - Y_1}{Y_0} \cdot 100$$

By taking the maximum values of the acceleration from data obtained using the software function we get $Y_0 \approx 8.57 \times 10^{-3} \text{ m/s}^2$ and $Y_1 \approx 6.94 \times 10^{-3} \text{ m/s}^2$, the reduction is

$$\Delta Y\% = \frac{8.57 \cdot 10^{-3} - 6.94 \cdot 10^{-3}}{8.57 \cdot 10^{-3}} \cdot 100 \approx 19\%$$

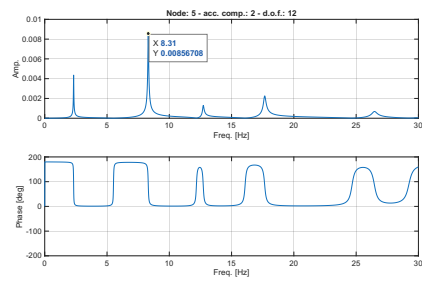


Figure 24: Figura 1

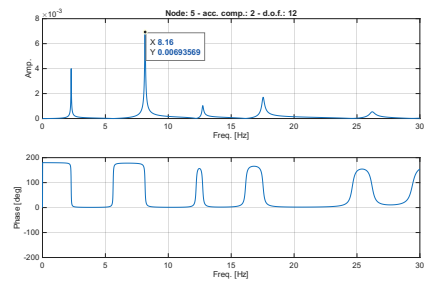


Figure 25: Figura 2